

Exercises on Pandas**Exercise 1: Creating a Series with Custom Index**

Create a pandas Series containing the populations of five cities (e.g., Istanbul: 15M, Ankara: 5M, Izmir: 4M, Bursa: 3M, Antalya: 2M). Ensure the city names are used as the index.

Exercise 2: Dictionary to DataFrame

Given the following dictionary representing company data, convert it into a pandas DataFrame:

```
data = {  
    'Project': ['SAYEM', 'TEYDEB', 'DPP', 'SmartAgri'],  
    'Year': [2024, 2025, 2024, 2026],  
    'Status': ['Completed', 'Pending', 'Active', 'Proposed']  
}
```

Exercise 3: Attribute Access and Indexing

From the DataFrame created in Exercise 2, retrieve only the 'Project' column using both attribute access (dot notation) and the loc operator.

Exercise 4: Reindexing a Series

Create a Series with values [10, 20, 30] and index ['a', 'c', 'e']. Use the reindex method to expand the index to ['a', 'b', 'c', 'd', 'e']. Use the ffill (forward fill) method to handle the missing values.

Exercise 5: Dropping Entries

Create a 3x3 DataFrame with random numbers and index/column labels. Demonstrate how to drop a specific row by label and then how to drop a specific column using the axis=1 argument.

Exercise 6: Boolean Indexing

Create a DataFrame of 10 rows containing a column 'Scores' with values between 0 and 100. Filter the DataFrame to show only the rows where the Score is greater than 70 and less than 90.

Exercise 7: Precise Selection with iloc

Using a 5x5 DataFrame, use iloc to select the sub-section comprising the 2nd and 3rd rows and the last three columns.

Exercise 8: Ranking and Sorting

Create a Series with some duplicate values and some NaN values. Sort the Series by its values (ensuring NaN appears at the end) and then calculate the rank of each element using the 'first' method to handle ties.

Exercise 9: Alignment of Differing Shapes

Create two DataFrames, df1 (3x2) and df2 (2x3), with overlapping but not identical row/column labels. Add them together and explain why certain cells result in NaN.

Exercise 10: Addition with Fill Values

Repeat Exercise 9, but use the .add() method with a fill_value=0 argument to ensure that non-overlapping cells do not result in NaN.

Exercise 11: Operations Between DataFrame and Series

Create a 4x4 DataFrame of random floats. Create a Series from the first row of that DataFrame. Subtract the Series from the DataFrame and observe how the operation broadcasts across the rows.

Exercise 12: Applying Functions with apply

Write a function that calculates the "range" (max - min) of a sequence. Apply this function to every column of a 5x5 DataFrame containing project budget data.

Exercise 13: Custom Element-wise Formatting

Create a DataFrame with floating-point numbers. Use `applymap` to format the entire DataFrame so that every number is displayed as a string with only two decimal places.

Exercise 14: Correlation and Covariance

Create a DataFrame representing stock prices for four companies (e.g., AAPL, IBM, MSFT, GOOG) over 5 days. Calculate the correlation matrix for the entire DataFrame. Then, create a separate Series representing trading volume and use `corrwith` to find the correlation between the stock prices and the volume.

Exercise 15: Membership and Unique Values

Given a Series with highly repetitive data (e.g., ['C', 'A', 'D', 'A', 'A', 'B', 'B', 'C', 'C']):

1. Extract the unique values.
2. Calculate the value counts.
3. Use the `isin` method to create a boolean mask for only 'B' and 'C' values and use that mask to filter the original Series.